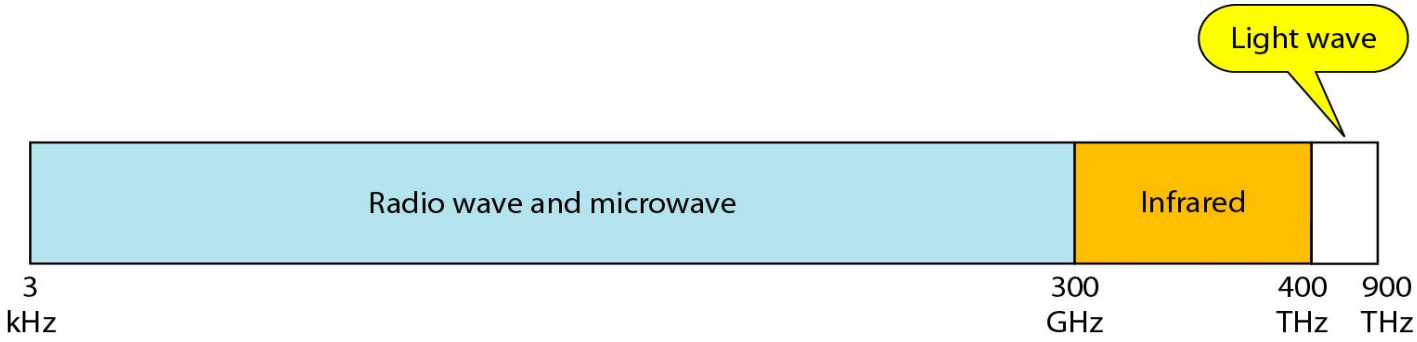
Data Communication
CSE 2211

Chapter 7 : Transmission Media (Unguided)

7-1 UNGUIDED MEDIA

- *Unguided media transport electromagnetic waves without using a physical conductor.*
- *This type of communication is often referred to as wireless communication.*
- *Signals are normally broadcast through free space and thus are available to anyone who has a device capable of receiving them.*
- *Electromagnetic spectrum for wireless communication:*

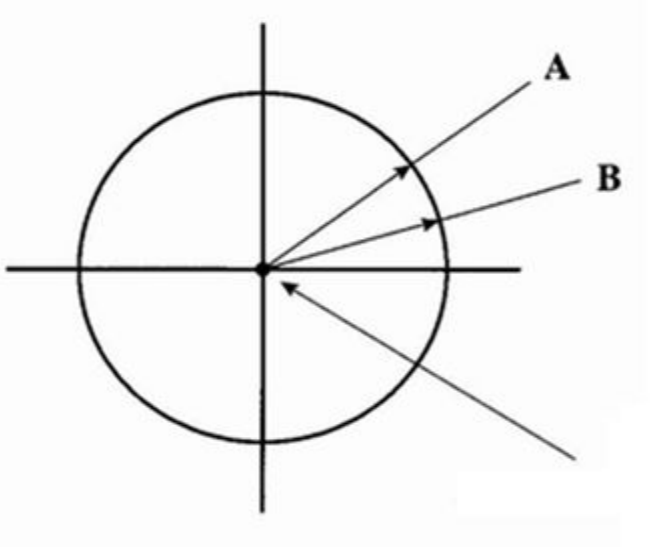


Antenna

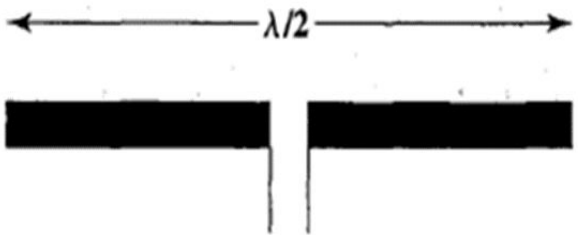
- An antenna is an electrical conductor or system of conductors
  - Transmission - radiates electromagnetic energy into space
  - Reception - collects electromagnetic energy from space
- In two-way communication, the same antenna can be used for transmission and reception

Types of Antennas

- **Isotropic antenna (idealized)**
  - Radiates power equally in all directions.
  - The pattern of radiation is sphere with the antenna at the center.

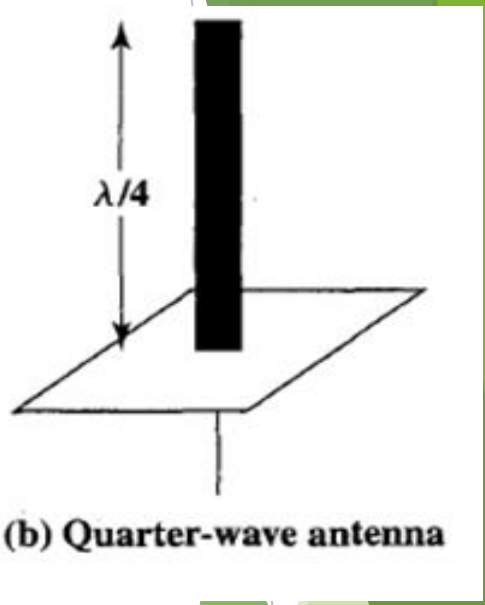


- **Dipole antennas**
  - Half-wave dipole antenna (or Hertz antenna):
    - An antenna having a physical length of ½ the wavelength is called a hertz antenna or a half wave dipole antenna.



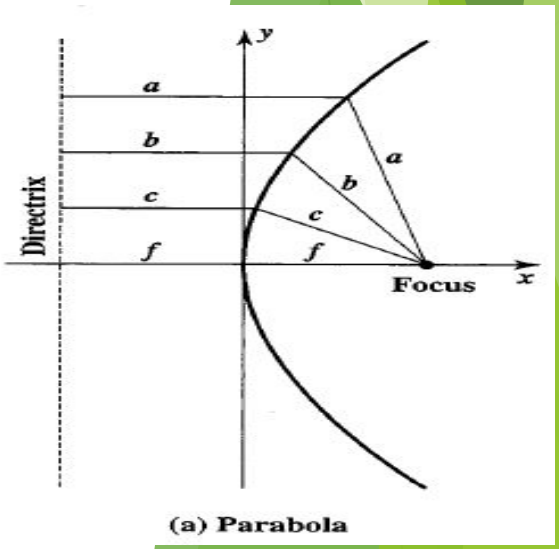
(a) Half-wave dipole

- **Dipole antennas**
  - Quarter-wave vertical antenna (or Marconi antenna) :
    - An antenna having a physical length of 1/4 the wavelength is called a Marconi antenna or a Quarter-wave vertical antenna.



(b) Quarter-wave antenna

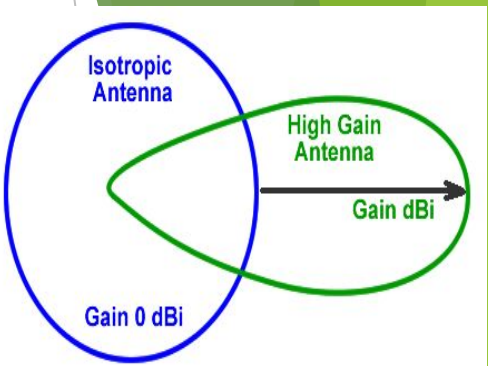
- **Parabolic Reflector Antenna**
  - Used in terrestrial microwave and satellite applications.



(a) Parabola

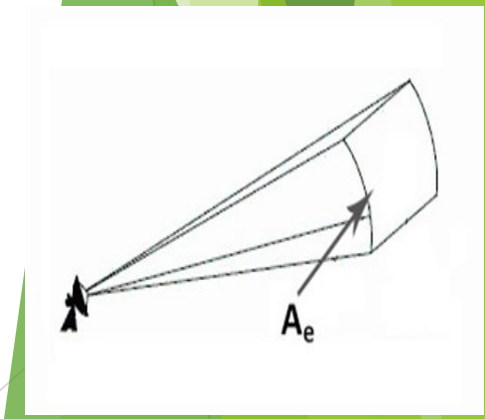
Antenna Gain

- Power output, in a particular direction, compared to that produced in any direction by a perfect omnidirectional antenna (isotropic antenna)
- G = (power radiated by an antenna)/(power radiated by reference antenna)



Effective Area of Antenna

- **Effective area (A_e)**
  - Related to physical size and shape of antenna
  - It is a theoretical value which is a measure of how effective an antenna is at receiving power.



- **Relationship between antenna gain and effective area**

$$G = \frac{4\pi A_e}{\lambda^2} = \frac{4\pi f^2 A_e}{c^2}$$

  - G = antenna gain
  - A_e = effective area
  - f = carrier frequency
  - c = speed of light (3×10^8 m/s)
  - λ = carrier wavelength

Effective Area and G for Different Types of Antennas

Type of Antenna	Effective Area A_e (m^2)	Power Gain (relative to isotropic)
Isotropic	$\lambda^2/4\pi$	1
Infinitesimal dipole or loop	$1.5\ \lambda^2/4\pi$	1.5
Half-wave dipole	$1.64\ \lambda^2/4\pi$	1.64
Horn, mouth area A	$0.81\ A$	$10\ A/\lambda^2$
Parabolic, face area A	$0.56\ A$	$7\ A/\lambda^2$
Turnstile (two crossed, perpendicular dipoles)	$1.15\ \lambda^2/4\pi$	1.15

Effective Area and G for Different Types of Antennas

Type of Antenna	Effective Area A_e (m^2)	Power Gain (relative to isotropic)
Isotropic	$\lambda^2/4\pi$	1
Infinitesimal dipole or loop	$1.5\ \lambda^2/4\pi$	1.5
Half-wave dipole	$1.64\ \lambda^2/4\pi$	1.64
Horn, mouth area A	$0.81\ A$	$10\ A/\lambda^2$
Parabolic, face area A	$0.56\ A$	$7\ A/\lambda^2$
Turnstile (two crossed, perpendicular dipoles)	$1.15\ \lambda^2/4\pi$	1.15

Example

1. For a parabolic reflective antenna with a diameter of 2 m, operating at 12 GHz, what is the effective area and gain?

Example 5.1 For a parabolic reflective antenna with a diameter of 2 m, operating at 12 GHz, what is the effective area and the antenna gain? We have an area of $A = \pi r^2 = \pi$ and an effective area of $A_e = 0.56\pi$. The wavelength is $\lambda = c/f = (3 \times 10^8)/(12 \times 10^9) = 0.025$ m. Then

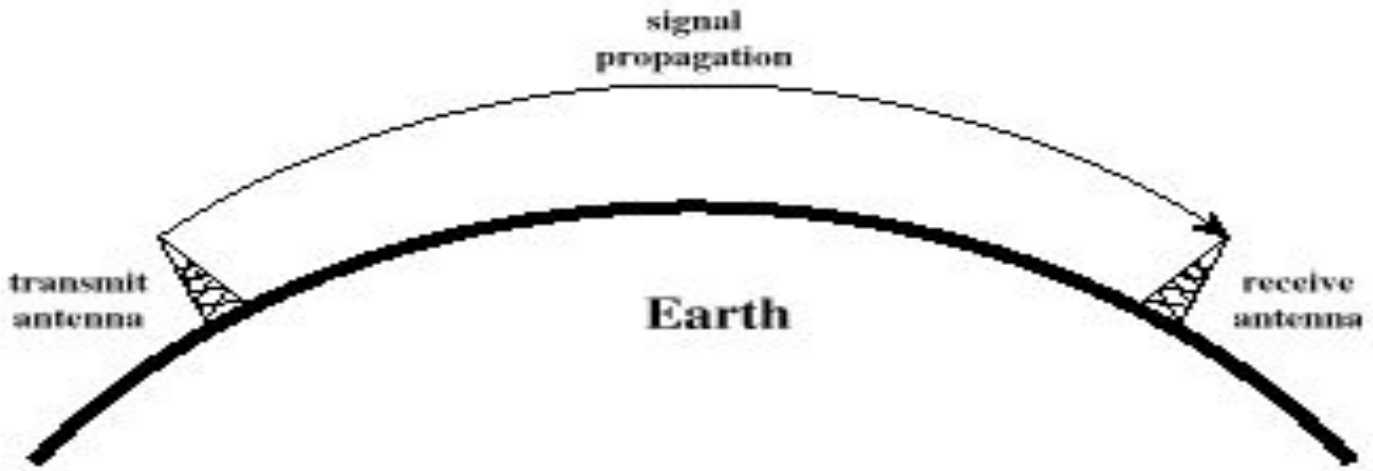
$$G = (7A)/\lambda^2 = (7 \times \pi)/(0.025)^2 = 35{,}186$$

$$G_{\pi B} = 45.46 \text{ dB}$$

Propagation Modes

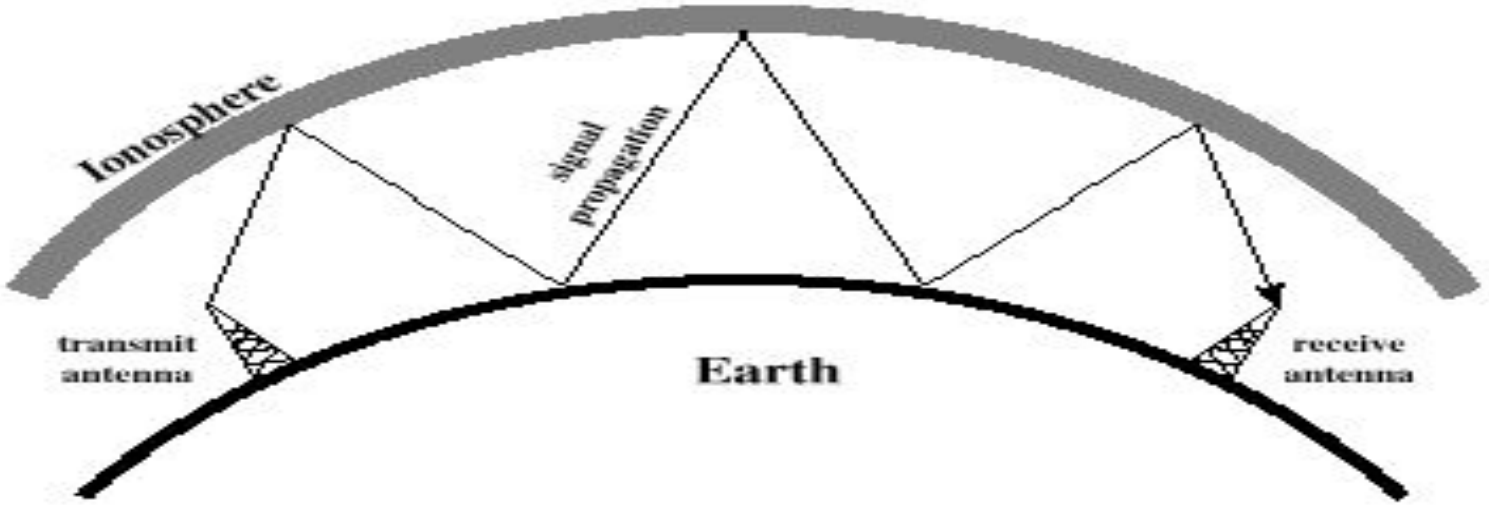
- Unguided signals can travel from the source to the destination in several ways:
  - Ground Wave (GW) Propagation: < 3MHz
  - Sky Wave (SW) Propagation: 3MHz to 30MHz
  - Effective Line-of-Sight (LOS) Propagation: > 30MHz

Ground Wave Propagation



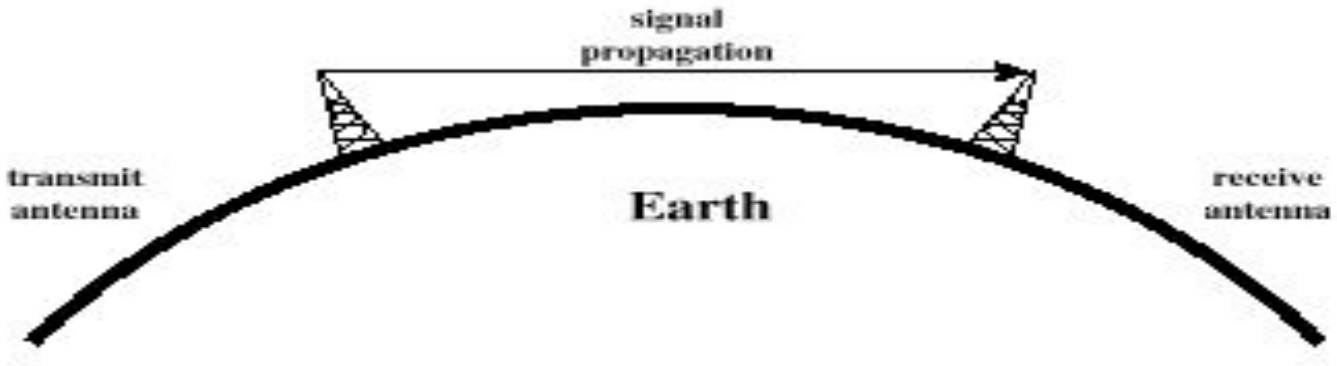
- Follows contour of the earth.
- Can propagate considerable distances.
- Frequency bands: ELF, VF, VLF, LF, MF.
- Spectrum range: 30Hz ~ 3MHz, e.g. AM radio.

Sky Wave Propagation



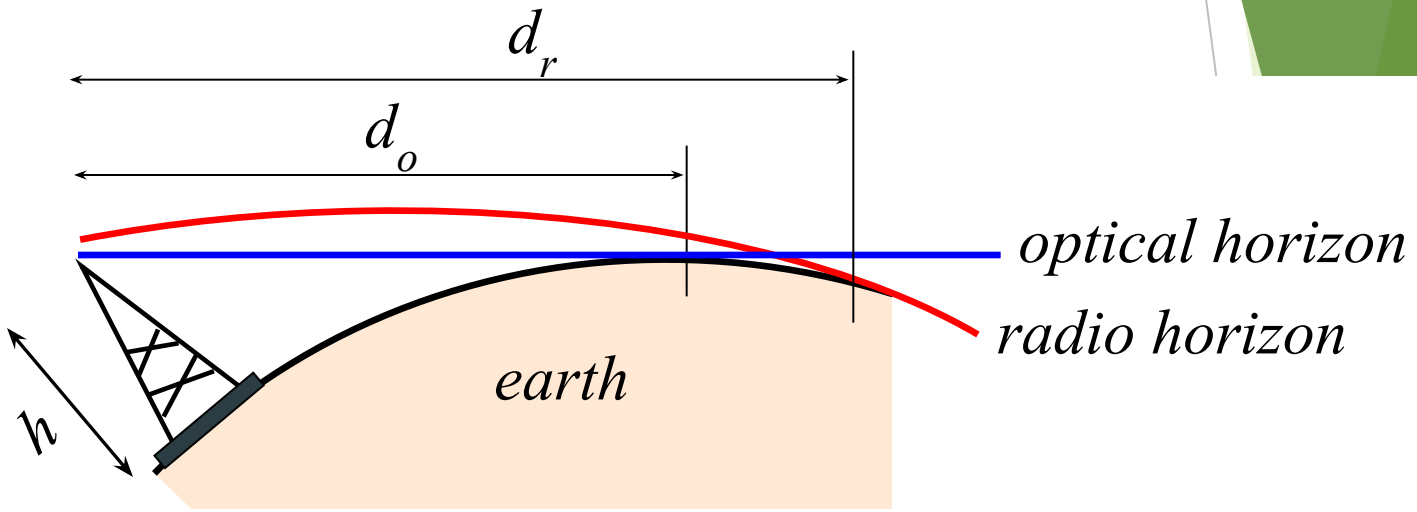
- Signal reflected from ionized layer of upper atmosphere back down to earth, which can travel a number of hops, back and forth between ionosphere and earth's surface.
- HF band with intermediate frequency range: 3MHz ~ 30MHz.
- Example International broadcast.

Line-of-Sight Propagation



- Above 30 MHz, neither ground wave nor sky wave propagation modes operate, and communication must be by line of sight (LOS)
- For satellite communication, signal above 30 MHz not reflected by ionosphere.
- For ground communication, antennas within *effective* LOS due to refraction.
- Frequency bands: VHF, UHF, SHF, EHF, Infrared, optical light
- Spectrum range : 30MHz ~ 900THz.

LOS calculations



- **What is the relationship between *h* and *d* ?**

– For optical LOS:

$$d_o = 3.57\sqrt{h}$$

– For effective or radio LOS:

$$d_r = 3.57\sqrt{Kh}$$

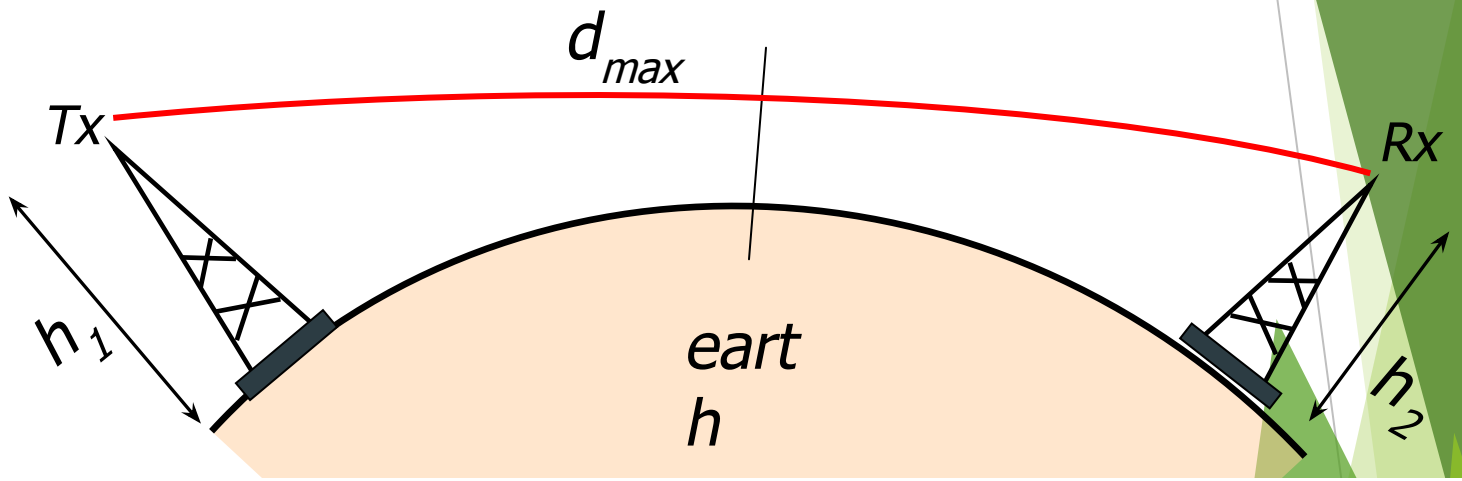
where

h = antenna height (m)

d = distance between antenna and horizon (km)

K = adjustment factor for refraction, K = 4/3

LOS calculations (2)



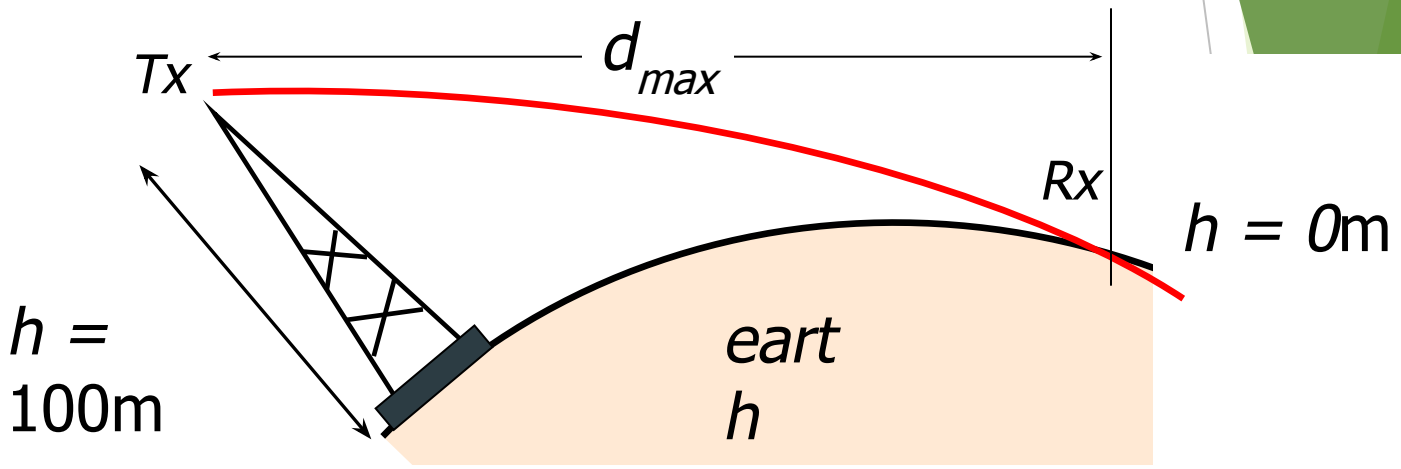
- **What is the Maximum distance between two antennas for LOS propagation ?**

$$d_{max} = 3.57\left(\sqrt{\mathrm{K}h_1} + \sqrt{\mathrm{K}h_2}\right)$$

where h_1 = height of antenna 1

h_2 = height of antenna 2

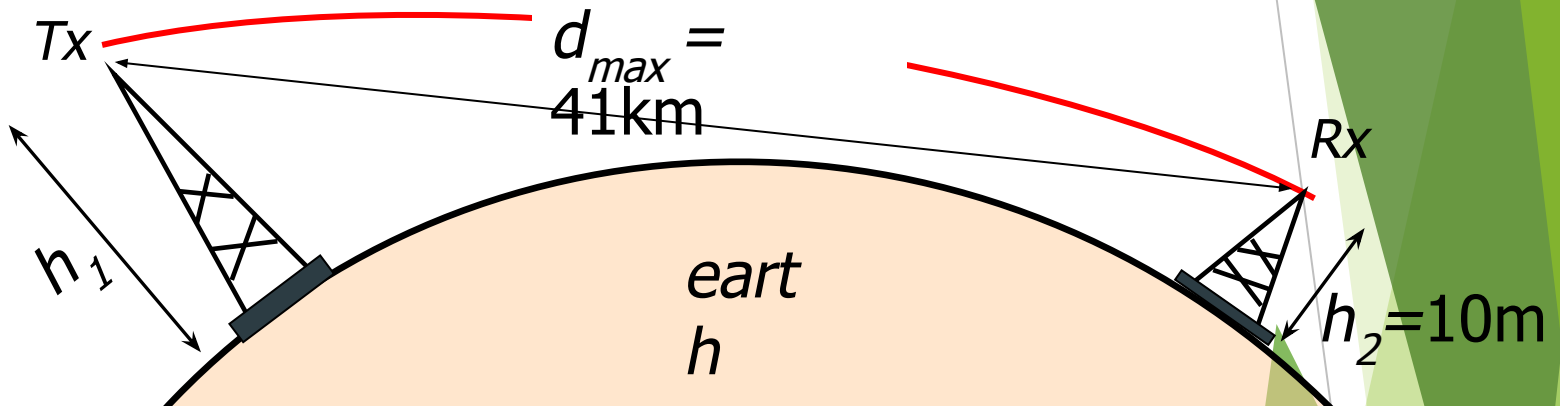
Example (1)



- What is the maximum distance between two antennas for LOS transmission if one antenna is 100m and the other is at ground level?

Solution: $d_{max} = 3.57\left(\sqrt{\mathrm{K}h_1} + 0\right) = 3.57\sqrt{\frac{4}{3}\times 100} = 41\mathrm{km}$

Example (2)



- Now, suppose the receiving antenna is 10m high, what is the height of the transmitting antenna in order to achieve same distance above?

$$d_{max} = 3.57\left(\sqrt{Kh_1} + \sqrt{Kh_2}\right) \Rightarrow 41 = 3.57\left(\sqrt{Kh_1} + \sqrt{\frac{4}{3}\times 10}\right)$$

$$\sqrt{Kh_1} = \frac{41}{3.57} - \sqrt{13.33} = 7.84$$

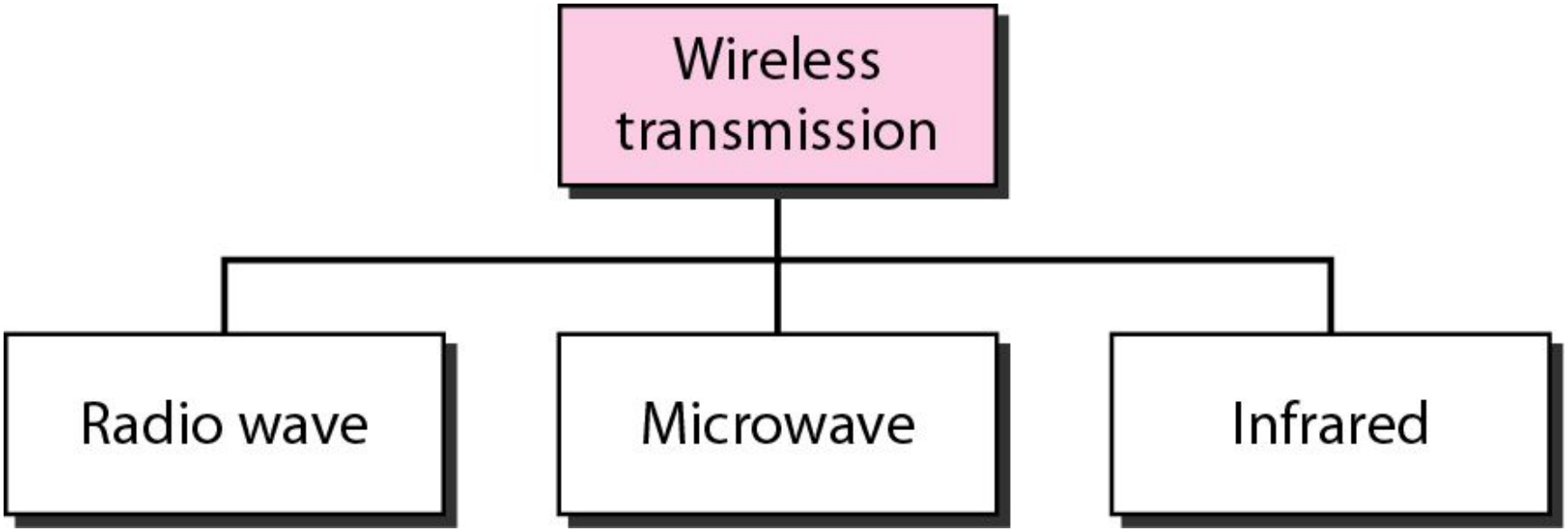
$$\therefore \quad h_1 = 7.84^2/1.33 = 46.2\text{m} \quad \text{(Saving over 50m)}$$

Problems

1. Find the optimum distance from the ground level for a half-wave dipole antenna of frequency 15 MHz.

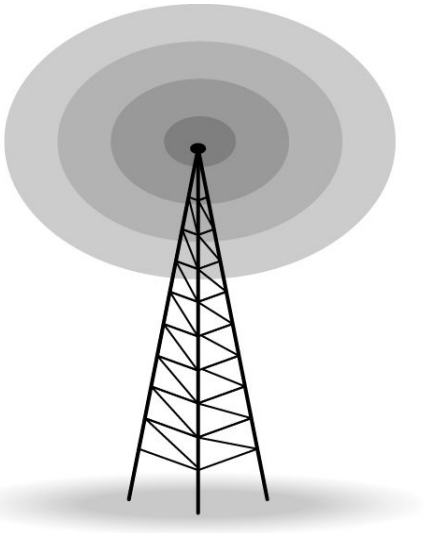
2. Determine the height of an antenna for a TV station that must be able to reach customers up to 80 km away.

Wireless Transmission Waves



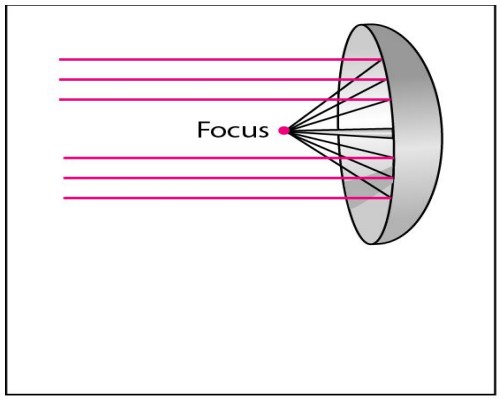
Radio Waves

- Electromagnetic waves ranging in frequencies between 3 kHz and 1 GHz are normally called **radio waves.**
- Use omnidirectional antennas that send out signals in all directions.
- Propagate in the sky mode, can travel long distances.
- Can penetrate through walls.
- Example- AM and FM radio, television.

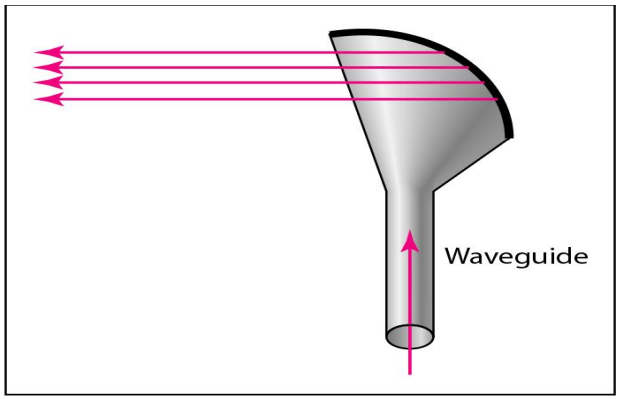


Micro Waves

- Electromagnetic waves having frequencies between 1 GHz and 300 GHz are called **microwaves.**
- Microwave propagation is line-of-sight.
- Very high-frequency microwaves cannot penetrate walls.
- Use unidirectional antennas that send out signals in one direction.
- Example- cellular phones, satellite networks, and wireless LANs



a. Dish antenna



b. Horn antenna

Infrared

- Infrared waves, with frequencies from 300 GHz to 400 THz (wavelengths from 1 mm to 770 nm), can be used for short-range communication.
- Having high frequencies, cannot penetrate walls.
- This advantageous characteristic prevents interference between one system and another.
- Infrared signals can be used for short-range communication in a closed area using line-of-sight propagation.